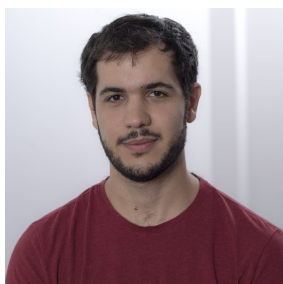


César Sabater - Currículum Vitae



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Nationality	Argentinian, Italian
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Email	cesar.sabater@insa-lyon.fr
Website	https://cesarsabater.github.io

Research Experience

March 2023 **Postdoctoral Researcher**
- present *Institution:* CNRS, INSA Lyon
Topic: Privacy and Security of Data-Processing Systems
Supervisor: Sonia Ben Mokhtar
Research Team: DRIM

July 2022 **Postdoctoral Researcher**
- March 2023 *Institution:* INRIA Lille - Nord Europe
Topic: Privacy-preserving machine learning
Supervisor: Jan Ramon
Research Team: MAGNET

Jan 2019 **Ph.D. in Computer Science**
- June 2022 *Institution:* INRIA Lille - Nord Europe
Defense date: June 20, 2022
Topic: Privacy-preserving machine learning
Supervisor: Jan Ramon
Research Team: MAGNET

Pre-Research Professional Experience

Jan - Dec 2018 **Research Engineer at INRIA Lille — MAGNET Team, France**
Topic: Development of cryptographic and graph-sampling tools
Supervisor: Jan Ramon

May - Nov 2014 **Internship at INRIA — CAMUS (Compilation for Multicore Architectures) Team, France**
Topic: Combining compiler optimization and numerical analysis for performance and productivity

Jan 2012 **Software Engineer at CIFASIS-CONICET, Argentina**
- Oct 2013 (International French-Argentine Center for Information Sciences and Systems)
Topic: Automatic metadata extraction and streamlined submission workflows for open-access academic repositories

July - Dec 2011 **Software Engineer at ITX Corp**
Co-founder and member of an independent Argentinian team of freelance developers working on assignments for ITX Corp., NY, USA
Main Tasks: Development of front-end mobile websites

July 2010 **Software Engineer at National University of Rosario, Argentina**
- Jun 2011 *Main Tasks:* Development of back-end and front-end extensions for online course platforms

Education

- Jan - March 2012 **Master Course Scholarship at IMPA (National Institute of Pure and Applied Mathematics)**
Rio de Janeiro, Brazil
*IMPA is a top-ranked institution dedicated to pure and applied mathematics.
There, I attended a master's-level course in extremal and probabilistic graph theory and combinatorics, taught by [Robert Morris](#).*
Grade: A-
- March 2008 **“Licenciado” Degree in Computer Science (Equivalent to a Master’s degree in France)**
- April 2017 *Qualification: Licenciado en Ciencias de la Computación*
University: Universidad Nacional de Rosario (National University of Rosario)
Academic Average: 9.47/10 (top 5% of the class)
*Although this degree is considered equivalent to a French BAC+5, it was completed over a longer period because it was pursued alongside several years of professional activity.
During this time, I held multiple software engineering positions in both the public and private sectors (see Pre-Research Professional Experience section).*

Publications

Conferences and Journals

- (SBR 2025) **Dropout-Robust Mechanisms for Differentially Private and Fully Decentralized Mean Estimation**
[César Sabater](#), Sonia Ben Mokhtar, and Jan Ramon
To appear in Proceedings on Privacy Enhancing Technologies (PoPETs) 2026
URL: <https://hal.science/hal-05308749>
- (NSM+ 2025) **PriviRec: Confidential and Decentralized Graph Filtering for Recommender Systems**
Julien Nicolas, [César Sabater](#), Mohamed Maouche, Mark Coates, and Sonia Ben Mokhtar
ACM International Conference on Information and Knowledge Management (CIKM), 2025
URL: <https://hal.science/hal-05308553/>
- (TJB+ 2025) **Exposing the Vulnerability of Decentralized Learning to Membership Inference Attacks Through the Lens of Graph Mixing**
Ousmane Touat, Jezekael Brunon, Yacine Belal, Julien Nicolas, [César Sabater](#), Mohamed Maouche, and Sonia Ben Mokhtar
ACM Middleware Conference, 2025
URL: <https://hal.science/hal-04933985v2>
- (SHP+ 2023) **Private Sampling with Identifiable Cheaters**
[César Sabater](#), Florian Hahn, Andreas Peter, and Jan Ramon
Proceedings on Privacy Enhancing Technologies (PoPETs) 2023
URL: <https://inria.hal.science/hal-03904200/>
- (SBR 2022) **An Accurate, Scalable and Verifiable Protocol for Federated Differentially Private Averaging**
[César Sabater](#), Aurélien Bellet, and Jan Ramon
Machine Learning Journal, 2022
URL: <https://inria.hal.science/hal-03820603>

Preprints

- (NSM+ 2024) **Differentially Private and Decentralized Randomized Power Method**
Julien Nicolas, [César Sabater](#), Mohamed Maouche, Sonia Ben Mokhtar, and Mark Coates
arXiv preprint, 2024
URL: <https://arxiv.org/pdf/2411.01931>

International Workshops

Zero Knowledge Arguments for Verifiable Sampling

César Sabater and Jan Ramon.

NeurIPS 2021 Workshop on Privacy in Machine Learning, 2021.

URL: <https://openreview.net/pdf?id=YS47RZFtKm>

Distributed Differentially Private Averaging with Improved Utility and Robustness to Malicious Parties

César Sabater, Aurélien Bellet, and Jan Ramon.

NeurIPS 2020 Workshop on Privacy Preserving Machine Learning (PriML & PPML Joint Edition) 2020

URL: <https://arxiv.org/abs/2006.07218>

Reviewing Service

- ICML 2023, 2024, 2025: Reviewer
- NeurIPS 2023, 2024, 2025: Reviewer. For NeurIPS 2023 I was ranked by area chairs among the top 8% of the reviewers.
- ICLR 2024, 2025: Reviewer
- AAAI 2024: PC Member
- ICDCS 2024: Reviewer
- ACM TOPS 2025: Reviewer
- IH&MMSec Workshop 2024: PC Member

Note that in the top Machine Learning conferences such as NeurIPS, ICML and ICLR, the role of reviewer is similar to the role of PC Member (which does not exist anymore in these venues) in conferences of similar rank on other fields (i.e., the reviewing load for reviewers in the above ML conferences is generally of 5 or 6 papers and reviewers engage in discussions with authors, other reviewers and area chairs).

Dissemination in Scientific Venues

- | | |
|---------------------------|--|
| July 7-11
2024 | Keynote Speaker at CYBER IN Occitanie 2025 Summer School in Cybersecurity
Font Romeu, France
https://cyberoccitanie.sciencesconf.org/ |
| June 24-27
2024 | Oral Presentation and Session Chair at 14ème Atelier sur la Protection de la Vie Privée (APVP 2024)
14th Privacy Protection Workshop
Lou Capitelle, France
https://apvp24.sciencesconf.org/ |
| Jan 29
-Feb 02
2024 | Session Chair at PEPR Cybersecurité Winter School 2024
Autrans, France
https://www.pepr-cybersecurite.fr/2023/11/10/ecole-d-hiver-2024/ |
| Sep 18-22
2023 | Oral Presentation at the 22th European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2023)
Torino, Italy
https://ecmlpkdd.org/2023/ |

- July 15-20 2023 **Oral presentation at the 24th Privacy Enhancing Technologies Symposium (PETs 2023)**
Lausanne, Switzerland
<https://petsymposium.org/2024/>
- March 9-10 2023 **Speaker at 2nd RedChain-Lab Workshop**
La Clusaz, France
<https://sites.google.com/view/red-chain-lab-laclusaz/>
- Jan 13 2023 **Speaker at Workshop de la chaire Data et Services pour une Ville Durable (DSVD)**
Lyon, France
<https://sites.google.com/view/workshop-dsvd-2023/accueil>
- June 23-24 2022 **Oral presentation at ANR Privacy-Preserving Methods for Medical Research (PMR) Workshop 2022**
Lille, France
<https://project.inria.fr/anrpmr/project-meeting-23-24-06-2022/>
- Dec 14 2021 **Poster and Recording at NeurIPS 2021 workshop on Privacy in Machine Learning (PriML)**
Virtual workshop
<https://priml2021.github.io/>
- Oct 27-28 2021 **Oral presentation at ANR Privacy-Preserving Methods for Medical Research (PMR) Workshop 2021**
Lyon, France
<https://project.inria.fr/anrpmr/project-meeting-27-10-2021/>
- December 11 2020 **Poster and Recording at NeurIPS 2020 workshop on Privacy Preserving Machine Learning**
PriML and PPML Joint Edition
Virtual workshop
<https://ppml-workshop.github.io/ppml20/>
- June 23-26 2020 **Recorded Presentation at cAP 2020 Conference**
Virtual workshop in France
<https://cap-rfiap2020.sciencesconf.org/>
- July 9-11 2019 **Poster Presentation at APVP 2019 — l'Atelier sur la Protection de la Vie Privée**
Cap Hornu, France
<https://project.inria.fr/apvp2019/>

Awards and Participation in National and International Competitions

National Competitions

- 2006 **5th position in National Selection Contest for IOI (International Olympiad in Informatics)**
1st substitute of the Argentinian team for the IOI that took place in Yucatan, México.
Buenos Aires, Argentina
- 2006 **Bronze Medal (3rd position) in Olimpiada Informática Argentina**
(Argentinian Olympiad in Informatics)
National Contest
Buenos Aires, Argentina

2004 5th position in Olimpiada Informática Argentina (Argentinian Olympiad in Informatics)
National Contest
Buenos Aires, Argentina

Engineering and Programming Skills

Advanced Knowledge in C, C++, STL, Analysis and Algorithm Design, and Data Structures. Advanced Knowledge in Java, PHP and Python. Advanced Knowledge Functional Programming in Haskell and OCaml.

Languages

French Advanced writing, reading, oral comprehension and speaking levels.
English Advanced writing, reading, oral comprehension and speaking levels.
Spanish Native Speaker.